

I will explore a research article by Engel et al. (2025) in response to Eric Siegel's essay, "To Fight Bias in Predictive Policing, Justice Can't be Color-Blind" published in "97 Things About Ethics Everyone in Data Science Should Know" (Franks 2020, p. 243-6). Siegel argues that data on race must be used to test "color blind" algorithms for racial bias. He claims risk assessment systems use proxies for race, leading to unequal outcomes. In this case, higher risk scores that lead to more false positive recommendations for detainment of black defendants compared to white defendants awaiting trial. Engel et al. (2025) test the algorithm mentioned in Siegel's piece using race-inclusive data to determine the extent of bias and apply machine learning to correct for that bias.

First, it will help to know the scope of the issue. According to Movement Alliance Project, a non-profit organization dedicated to ending pretrial detention, risk assessment algorithms are used in 11 states, and 178 "additional counties in other states" ("Where Are Risk Assessments Being Used?," n.d.). Assuming those numbers are correct, Siegel's observation of the ProPublica report that COMPAS, a risk assessment system, "falsely flags white defendants at a rate of 23.5% and black defendants at a rate of 44.9%" is not only concerning because of the disparity, but that it may be widespread.

Next, it will help to know the extent to which these systems are relied upon. In the Engel et al., (2025) paper, the authors observe the producer of COMPAS "immunizes itself from criticism" by noting the decision to detain is left to a "competent judge" who makes "the choice between bail and jail" (p. 385). Therefore, it is important to inquire how often judges cite COMPAS as the reason for detaining a defendant. According to Perplexity AI ([link to transcript](#)), studies show judges often override recommendations by risk assessment software. When prompted to provide examples of research, Perplexity AI concluded that judges have been found to rely on COMPAS recommendations, but frequency and degree could not be determined. Thus, judges are a third variable that may explain the disparity in false positives between black and white defendants.

Engel et al. (2025) argue that even if these systems were optimized to reduce errors, they still violate the tenants of democracy. They observed the overall tendency of COMPAS was to jail defendants who would not have recidivated over releasing defendants who would, thus violating the "'clear and convincing evidence' test" (p. 386), which required the evidence be substantially more likely to be true than untrue (p. 384). Because COMPAS is produced by a "commercial entity," and the algorithm's process is "concealed from public scrutiny," the process "lacks democratic legitimacy" (p. 386). Finally, even when the authors trained their own algorithm to correct for bias, bias was reduced overall but increased in the higher decile scores (p. 391).

My takeaway from this is that the public must have a say in whether risk assessment software is used. When it is used, the algorithm must be transparent, and it must be tested regularly.

References

- Engel, C., Linhardt, L., & Schubert, M. (2025). Code is law: How COMPAS affects the way the judiciary handles the risk of recidivism. *Artificial Intelligence and Law*, 33(2), 383–404. <https://doi.org/10.1007/s10506-024-09389-8>
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